

CPEN 416 / EECE 5710

Lecture 08: The oracle, query complexity, and Deutsch's algorithm

Tuesday 7 October 2025

Announcements

- No quiz today
- Midterms available for pickup after class
- Project details coming soon (start thinking about groups)

Module 2 learning outcomes

Learning outcomes:

- explain what it means for an algorithm to have a quantum speedup
- define quantum oracles and query complexity
- implement oracles and Grover's algorithm in PennyLane
- identify the different components of the quantum compilation stack
- define and list common universal gate sets
- estimate the resources required to run a quantum algorithm
- perform simple circuit optimizations in PennyLane

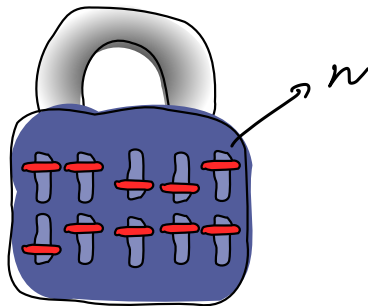
Today

Learning outcomes:

- Define the query complexity of an algorithm
- Describe multiple strategies for implementing an *oracle* query with a quantum circuit
- Apply the phase kickback trick
- Implement Deutsch's algorithm to determine whether a Boolean function is constant or balanced (in theory and in PennyLane)

Oracles: motivating problem

Suppose we would like to find the combination for a “binary” lock:



How do we solve this classically?

2^n tries
(worst case)

Image credit: Codebook node A.1

Idea: use superposition

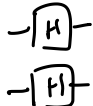
Can we do better with a quantum computer?

Idea: take n qubits and construct a uniform superposition of all combinations

$$\frac{1}{\sqrt{2}^n} (|0-0\rangle + |0-01\rangle + \dots + |11\dots 10\rangle + |11\dots 1\rangle)$$

Exercise: design a circuit to create this state.

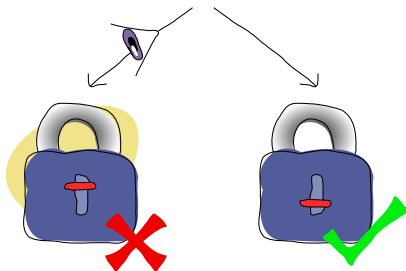
1 qubit: 

2 qubits: 

n qubits: 

Idea: use superposition

Measurements are probabilistic - the solution just being “in” the superposition doesn’t make it easier to solve the problem.



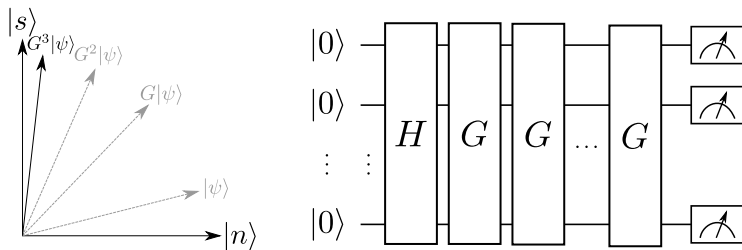
Quantum computers are **NOT** faster because they “compute everything at the same time.”

Image credit: Codebook node A.1

Solving problems with quantum computers

Can we solve this problem better with a quantum computer?

Yes: **amplitude amplification**, and **Grover's algorithm**



We will explore the algorithmic primitives, and see other cases where quantum computers have a **speed up in query complexity**.

Oracles

Revisit the lock problem and model a “try” as a function. Let

- $\mathbf{x}$ be an n -bit string (an input combination)
- $\mathbf{s}$ be the correct combination

$$f(\vec{x}) = \begin{cases} 1 & \vec{x} = \vec{s} \\ 0 & \text{otherwise} \end{cases}$$



Don't care *how* $f(\mathbf{x})$ is evaluated, only that it gives a “yes/no” answer. $f(\mathbf{x})$ is a black box, or **oracle**.

Every time we try a combination, we are **querying the oracle**.
The amount of queries is the **query complexity**.

classical case: $O(2^n)$
worst

Quantum oracles

Must determine

- a mathematical description of an oracle
- where to record the output, i.e., the value of $f(x)$

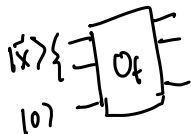
$$|\vec{x}\rangle \left\{ \begin{array}{c} \boxed{?} \\ \vdots \\ \boxed{f(\vec{x})} \end{array} \right\} = |\vec{x}\rangle \otimes |f(\vec{x})\rangle$$

Idea 1: encode the result in the state of an additional qubit.

$$O_f(|\vec{x}\rangle|y\rangle) = |\vec{x}\rangle|y \oplus f(\vec{x})\rangle$$

input answer

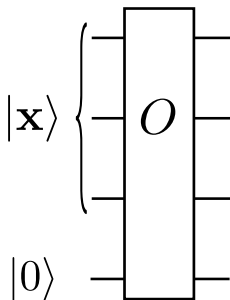
Exercise: Consider a 2-qubit system where $f(01) = 1$, and $f(x) = 0$ for all other x . What is the action of the oracle?



Input state	Output state
$ 00\rangle 0\rangle$	$ 00\rangle 0 \oplus f(00)\rangle = 00\rangle 0\rangle$
$ 01\rangle 0\rangle$	$ 01\rangle 0 \oplus f(01)\rangle = 01\rangle 1\rangle$
$ 10\rangle 0\rangle$	$ 10\rangle 0 \oplus f(10)\rangle = 10\rangle 0\rangle$
$ 11\rangle 0\rangle$	$ 11\rangle 0 \oplus f(11)\rangle = 11\rangle 0\rangle$



Quantum oracles



$$O|000\rangle|0\rangle = |000\rangle|0\rangle$$

$$O|001\rangle|0\rangle = |001\rangle|0\rangle$$

$$O|010\rangle|0\rangle = |010\rangle|0\rangle$$

$$O|011\rangle|0\rangle = |011\rangle|0\rangle$$

$$O|100\rangle|0\rangle = |100\rangle|0\rangle$$

$$O|101\rangle|0\rangle = |101\rangle|0\rangle$$

$$O|110\rangle|0\rangle = |110\rangle|1\rangle$$

$$O|111\rangle|0\rangle = |111\rangle|0\rangle$$

Quantum oracles

$$|\vec{x}\rangle \equiv \boxed{0_f}$$

Idea 2: encode the result in the phase

$$\frac{1}{\sqrt{2}}(|100\rangle + |010\rangle)$$

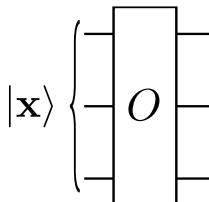
$$O|\vec{x}\rangle = (-1)^{f(\vec{x})}|\vec{x}\rangle$$

Exercise: Consider a 2-qubit system where $f(11) = 1$, and $f(\mathbf{x}) = 0$ for all other $\mathbf{x}$. What is the action of the oracle?

Input state	Output state
$ 00\rangle$	$ 00\rangle$
$ 01\rangle$	$ 01\rangle$
$ 10\rangle$	$ 10\rangle$
$ 11\rangle$	$(-1) \cdot 11\rangle$



Quantum oracles



$$O|000\rangle = |000\rangle$$

$$O|001\rangle = |001\rangle$$

$$O|010\rangle = |010\rangle$$

$$O|011\rangle = |011\rangle$$

$$O|100\rangle = |100\rangle$$

$$O|101\rangle = |101\rangle$$

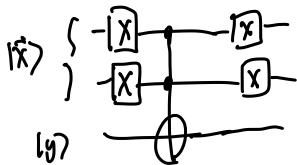
$$O|110\rangle = -|110\rangle$$

$$O|111\rangle = |111\rangle$$

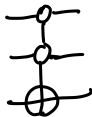
Oracle circuits

Exercise: develop two quantum circuit implementations for a 2-input oracle with function $f(00) = 1$.

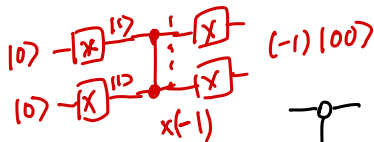
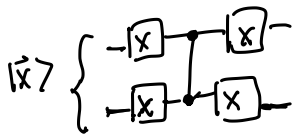
"type 1" (qubit)



OR



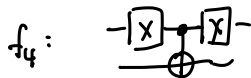
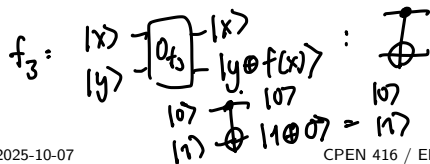
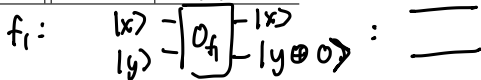
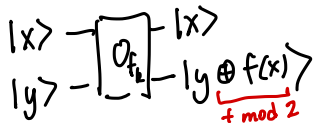
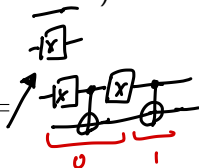
"type 2" (phase)



Oracle circuits

Exercise: construct two-qubit oracle circuits ("type 1" oracle) for the following four functions.

Name	Action	Name	Action
f_1	$f_1(0) = 0$ $f_1(1) = 0$	f_2	$f_2(0) = 1$ $f_2(1) = 1$
f_3	$f_3(0) = 0$ $f_3(1) = 1$	f_4	$f_4(0) = 1$ $f_4(1) = 0$



Deutsch's algorithm

Motivation: You have access to an oracle that implements one of the following functions:

Name	Action	Name	Action
f_1	$f_1(0) = 0$	f_2	$f_2(0) = 1$
	$f_1(1) = 0$		$f_2(1) = 1$
f_3	$f_3(0) = 0$	f_4	$f_4(0) = 1$
	$f_3(1) = 1$		$f_4(1) = 0$

Functions f_1 and f_2 are *constant*, f_3 and f_4 are *balanced*.

→ 2

How many **classical** oracle queries to determine if the oracle function is constant or balanced? How many **quantum** queries?

↓
1

Phase kickback

The secret is the *phase kickback* trick.

Exercise: apply CNOT_{01} to the following two-qubit states.

$$|0\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right), \quad |1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$



$$|1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |1\rangle \left(\frac{|1\rangle - |0\rangle}{\sqrt{2}} \right) = |1\rangle (-1) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = -|1\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$|0\rangle |-\rangle \xrightarrow{\text{CNOT}} |0\rangle |-\rangle$$

$$|1\rangle |-\rangle \rightarrow -|1\rangle |-\rangle$$

$$|b\rangle |-\rangle \rightarrow (-1)^b |b\rangle |-\rangle$$

Deutsch's algorithm

Let U_f be an oracle for one of the four functions:

$$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

Initializing the second qubit to $|-\rangle$ allows us to learn $f(0) \oplus f(1)$ with a single oracle query.

Q: How does $f(0) \oplus f(1)$ relate to properties of the function?

$$f(0) \oplus f(1) = \begin{array}{ll} 0 & \text{const.} \\ 1 & \text{balanced.} \end{array}$$

Deutsch's algorithm

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

$$\begin{aligned} U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) &= \frac{1}{\sqrt{2}} \left[U_f |x\rangle |0\rangle - U_f |x\rangle |1\rangle \right] \\ &= \frac{1}{\sqrt{2}} \left[|x\rangle |0 \oplus f(x)\rangle - |x\rangle |1 \oplus f(x)\rangle \right] \end{aligned}$$

If $f(x) = 0$,

$$U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |x\rangle |-\rangle$$

If $f(x) = 1$,

$$U_f |x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = (-1) |x\rangle |-\rangle$$

Deutsch's algorithm

Remember how we generalized the result for CNOT:

$$\text{CNOT} \left(|b\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \right) = (-1)^b |b\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right),$$

So we can write

$$U_f(|x\rangle|-\rangle) = (-1)^{f(x)}|x\rangle|-\rangle$$

Deutsch's algorithm

How to use this to get $f(0) \oplus f(1)$?

$$\begin{aligned}
 & U_f \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \\
 &= U_f \frac{|0\rangle}{\sqrt{2}} |-\rangle + U_f \frac{|1\rangle}{\sqrt{2}} |-\rangle \\
 &= \underline{(-1)^{f(0)}} \frac{|0\rangle}{\sqrt{2}} |-\rangle + \underline{(-1)^{f(1)}} \frac{|1\rangle}{\sqrt{2}} |-\rangle \\
 &= \left[(-1)^{f(0)} \frac{|0\rangle}{\sqrt{2}} + (-1)^{f(1)} \frac{|1\rangle}{\sqrt{2}} \right] \otimes |-\rangle \\
 &= (-1)^{f(0)} \left[\frac{|0\rangle}{\sqrt{2}} + (-1)^{f(0) \oplus f(1)} \frac{|1\rangle}{\sqrt{2}} \right] \otimes |-\rangle
 \end{aligned}$$

Deutsch's algorithm

$$U_f \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) \frac{|0\rangle + (-1)^{f(0) \oplus f(1)} |1\rangle}{\sqrt{2}} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

If the function is constant, $f(0) \oplus f(1) = 0$ and the state is

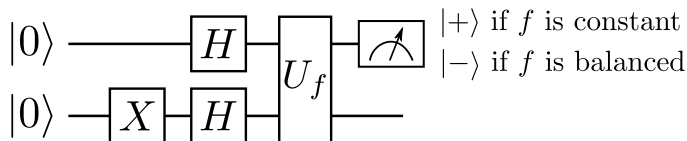
$$U_f \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |+\rangle |-\rangle$$

But if the function is balanced, $f(0) \oplus f(1) = 1$ and the state is

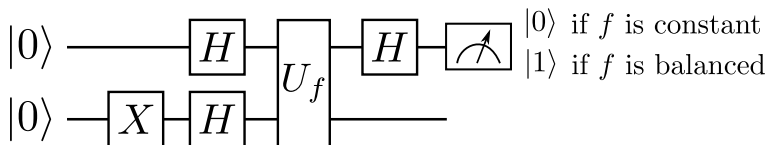
$$U_f \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |-\rangle |-\rangle$$

Deutsch's algorithm

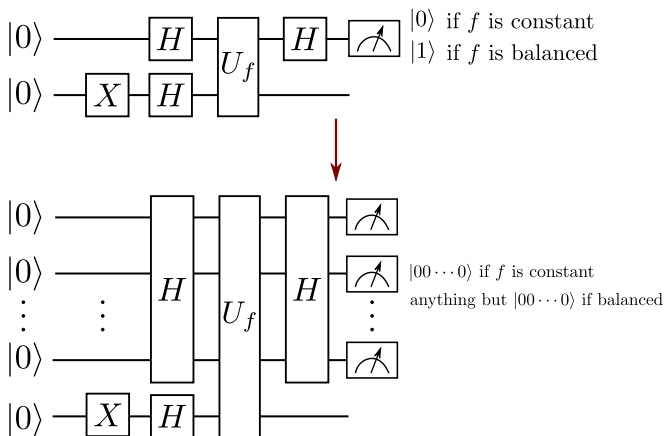
Deutsch's algorithm as a circuit:



Equivalently,



Generalization: Deutsch-Jozsa algorithm



$2^{n-1} + 1$ classical queries in worst case; still only 1 quantum query.
(Challenge: try implementing it yourself to check if this works!)

Oracle-based algorithms

A few other interesting algorithms:

Bernstein-Vazirani algorithm (practice A2)

Given $f : \{0, 1\}^n \rightarrow \{0, 1\}$ such that $f(x) = x \cdot s$ for some secret bitstring s . Find s using the fewest number of oracle queries.

Simon's algorithm

Given $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ and promised that for some non-trivial bit string s , $f(x) = f(y)$ iff $x \oplus y = s$. Find s using the fewest number of oracle queries.

Next time

Content:

- Amplitude amplification and Grover's algorithm

Action items:

- 1 Start thinking about project groups

Recommended reading:

- For today: Codebook module BA; Nielsen & Chuang 1.4.1-1.4.4
- For next class: Codebook module GA; Nielsen & Chuang 6.1